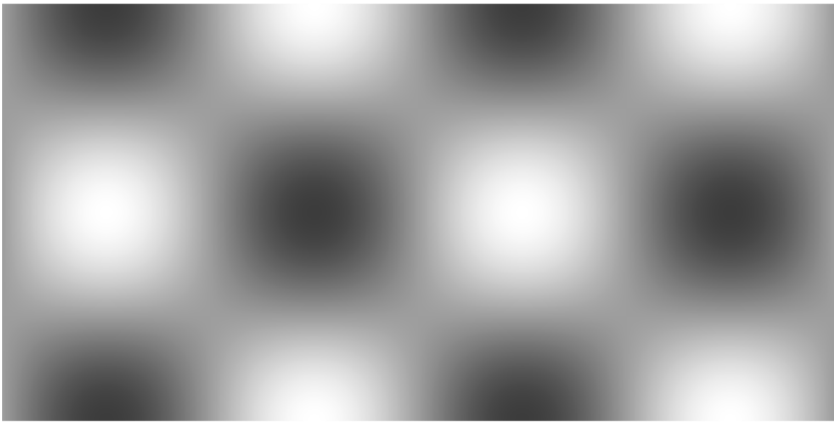
MATHEMATICAL READER / 01

A small book of uncertainty

From a probability to an equation, a curve and an image



CONTENTS

Contents

Two outcomes, one probability 3

A field of probabilities 7

Reading a curve without colour 9

What a threshold keeps 13

Sources and colophon 15

About this book

This is an expository typesetting specimen. Its numerical values and raster artwork are calculated from the functions printed here. They are not observations of people, experiments or model performance. The colour plates, grayscale analysis and monochrome appendix offer different views of those same calculations.

Chapter 1

Two outcomes, one probability

Imagine a source that can emit either zero or one. Let p be the probability of one; the probability of zero is then 1 − p. Naming two outcomes tells us what can happen, but not how surprising either outcome will be. A source that almost always emits zero behaves differently from one that chooses its two outcomes with equal probability.

Shannon’s entropy describes a probability distribution by the average information associated with its outcomes. With logarithms to base two, the unit is the bit. For a binary source the definition reduces to two terms, one for each possible outcome. [1]

$$H(p) = -p\log_2 p - (1-p)\log_2 (1-p) \tag{1}$$

At either endpoint, one outcome is certain. The convention $0 \log_2 0 = 0$ extends the formula continuously and gives $H(0) = H(1) = 0$. When both outcomes are equally probable, substituting $p = 1/2$ gives $H(1/2) = 1$ bit. The label “bit” belongs to the information measure, not to the probability itself.

Swapping the names zero and one changes p to $1 - p$. It changes neither the distribution of information nor the entropy. Consequently, the curve is symmetric about $p = 1/2$. A plot that lacks that symmetry would contradict the formula, even if the axes and typography looked convincing.

Knowing the probability is an assumption of this calculation. If p were estimated from a finite sample, we would also need to discuss that estimate and its uncertainty. The smooth curve here makes no statement about sample size, dependence between symbols or the behaviour of a particular compression program.

The following chapters separate three ways of seeing the same mathematical object. An image gives an overview of many probabilities arranged in space. A curve shows how a measure changes

as its input changes. A table preserves selected values at an explicit precision. Each answers a different reading task.

Chapter 2

A field of probabilities

A formula can also provide the pixels of an image. On the unit square, define a probability at every position (x, y) by

$$p(x, y) = \frac{1 + \sin(4\pi x)\cos(2\pi y)}{2} \tag{2}$$

Sine and cosine each lie between −1 and 1. Their product therefore stays in that interval, and the affine transformation maps it into [0, 1]. This is a constructed probability field, not a photograph or a measurement.

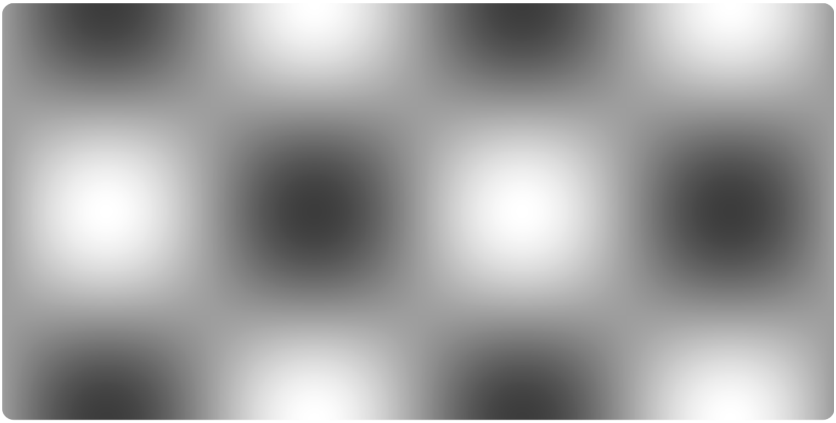


Figure 1. The probability field on $0 \le x, y \le 1$. White represents $p = 0$ and the darkest tone represents $p = 1$; x increases to the right and y upward.

Source: equation (2), sampled on a 1152×576 grid including its endpoints. RGB is interpolated from (255, 255, 255) to (0, 72, 112).

Chapter 3

Reading a curve without colour

Colour is only one way to distinguish quantities. In Figure 2, entropy H uses a solid line and filled circles. The comparison V uses a dashed line and hollow squares. The identity survives when both curves are printed with neutral inks.

Define a normalized variance for comparison:

$$V(p) = 4p(1 - p) \tag{3}$$

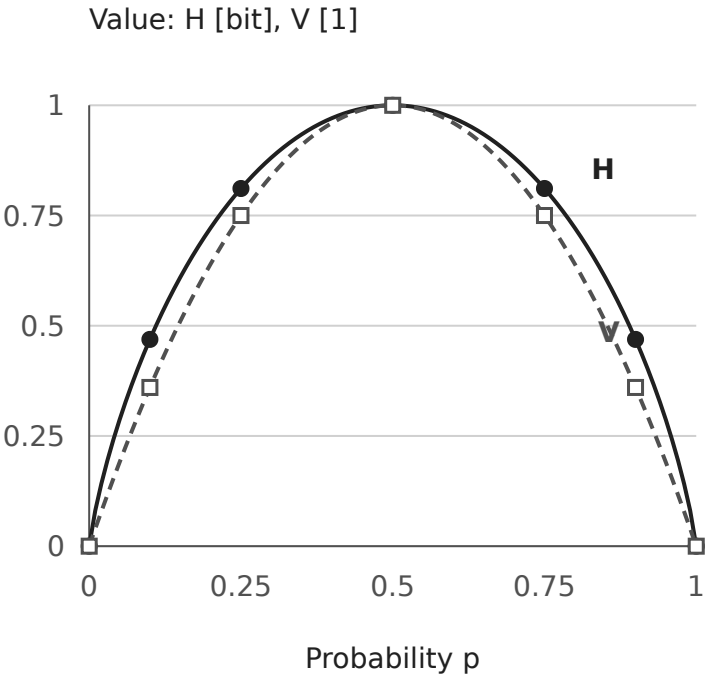


Figure 2. Entropy H in bits and normalized variance V without units. Both reach one at p = 1/2. Lines are evaluated at increments of 0.01; the symbols identify the tabulated probabilities.

Table 1. Values calculated from equations (1) and (3), rounded to three decimal places for display.

Probability p	Entropy H [bit]	Variance V [1]
0.00	0.000	0.000
0.10	0.469	0.360
0.25	0.811	0.750
0.50	1.000	1.000
0.75	0.811	0.750
0.90	0.469	0.360
1.00	0.000	0.000

At p = 0.1, H is approximately 0.469 bit and V is 0.360. Their endpoints and maxima agree, but their intermediate values do not. A common numerical range does not make two quantities interchangeable. Keep the units next to the values when comparing them.

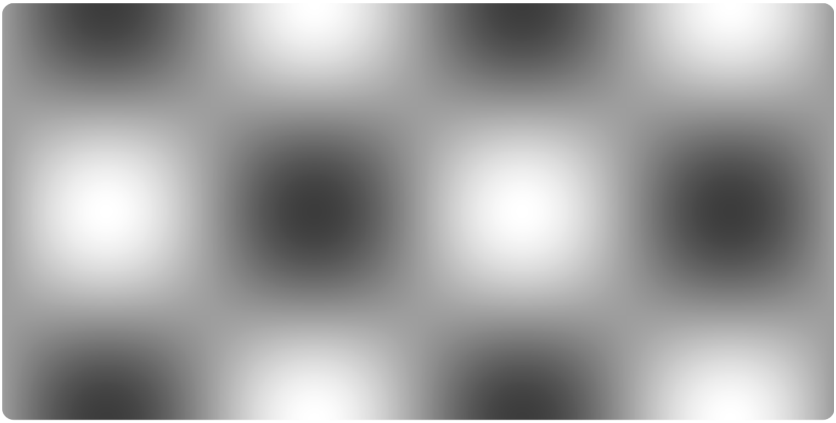


Figure 3. The same raster as Figure 1, rendered in grayscale. The spatial pattern remains, while the blue hue is removed.

Chapter 4

What a threshold keeps

A two-ink image must discard or encode its intermediate tones. Here the image is recalculated as black wherever p ≥ 1/2 and white elsewhere. The decision boundary remains; the magnitude within each region is lost.

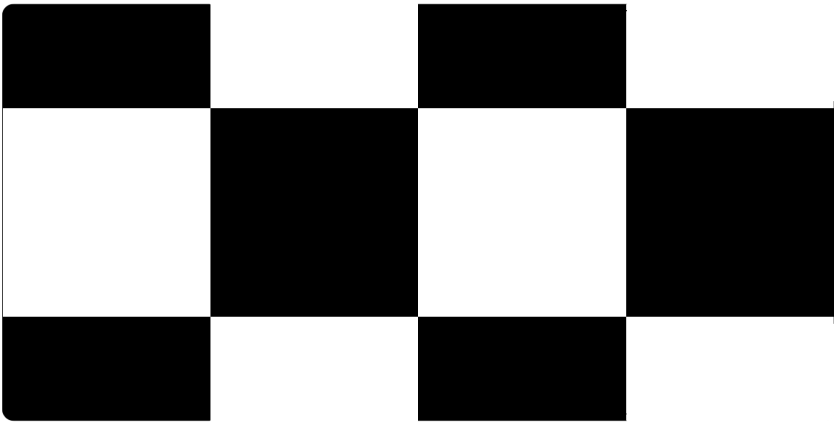


Figure 4. Thresholded probability field. Black means p ≥ 1/2; white means p < 1/2. This local PNG contains only black and white source pixels.

A dark region no longer tells us whether the probability was just above one half or close to one. This loss is part of the representation, so the caption states the threshold. The appendix uses the same distinction in its text, rules and illustration: black ink, white paper, explicit labels.

A picture does not remove the need for a definition. Equations specify the quantities, captions explain the encoding, and tables preserve values that can be compared without judging a shade or tracing a curve.

Chapter 5

Sources and colophon

[1] C. E. Shannon. A Mathematical Theory of Communication.
https://web.mit.edu/6.976/www/handout/shannon.pdf
The Bell System Technical Journal, 27, 379–423, 623–656 (1948).
The definition of entropy. The probability field, normalized variance comparison, calculations and illustrations in this book are original expository examples.

Typeset with doc-ui. Native MathML, SVG curves and local PNG artwork share the reading layout. Text stays selectable. The HTML can be read and printed without scripts or network access.